Data Communication & Computer Network

Dr.Sarang Patil

Associate Professor

Dept. of CSE,ASET

Unit-IV: Network Layer - IP Addressing

Contents: Network Layer :Network Layer Services, IPv4 Addresses: Classful and Classless Addressing, Special Addresses, NAT, Subnetting, Supernetting, Delivery and Forwarding of IP Packet, Structure of Router, IPv4: Datagrams, Fragmentation, Options, Checksum, Security, IPsec, IPv6Addressing: Notations, Address Space, Packet Format, Transition from Ipv4 to IPv6

Unit Objectives and outcomes: *On completion the students will be able to:*

1.To understand IPV 4 addressing

Unit outcomes **:** On completion the students will be able to
CO4: Apply the skills of subnetting, supernetting and routing mechanisms.
CO5: Differentiate IPv4 and IPv6.

.

Outcome Mapping:

PEOs: POs:1,2,4 COs: 4,5 PSOs:2

Books used

Simon Haykin ,” Communication Systems” , 3E

Ranjan Bose, ―Information Theory coding and Cryptography‖, McGraw-Hill, 2nd Ed

Bernad Sklar, ―Digital Communication Fundamentals & applications‖, Pearson Education. Second Edition.

Introduction to Network Layer

- **Network layer** is the third layer in the OSI model of computer networks.

- It's main function is to transfer network packets from the source to the destination.

- It is involved both at the source host and the destination host.

- At the source, it accepts a packet from the transport layer, encapsulates it in a datagram and then deliver the packet to the data link layer so that it can further be sent to the receiver.

- At the destination, the datagram is decapsulated, the packet is extracted and delivered to the corresponding transport layer.

Features :

- Main responsibility of Network layer is to carry the data packets from the source to the destination without changing or using it.

- If the packets are too large for delivery, they are fragmented i.e., broken down into smaller packets.

- It decides the root to be taken by the packets to travel from the source to the destination among the multiple roots available in a network (also called as routing).

- The source and destination addresses are added to the data packets inside the network layer.

The main functions performed by the network layer are:

Routing: When a packet reaches the router's input link, the router will move the packets to the router's output link. For example, a packet from S1 to R1 must be forwarded to the next router on the path to S2.

Logical Addressing: The data link layer implements the physical addressing and network layer implements the logical addressing. Logical addressing is also used to distinguish between source and destination system. The network layer adds a header to the packet which includes the logical addresses of both the sender and the receiver.

Internetworking: This is the main role of the network layer that it provides the logical connection between different types of networks.

Fragmentation: The fragmentation is a process of breaking the packets into the smallest individual data units that travel through different networks.

The services which are offered by the network layer protocol are as follows:

Packetizing

The process of encapsulating the data received from upper layers of the network(also called as payload) in a network layer packet at the source and decapsulating the payload from the network layer packet at the destination is known as packetizing.

Routing and Forwarding –

These are two other services offered by the network layer. In a network, there are a number of roots available from the source to the destination. The network layer specifies has some strategies which find out the best possible route. This process is referred to as routing.

Guaranteed delivery: This layer provides the service which guarantees that the packet will arrive at its destination.

Guaranteed delivery with bounded delay: This service guarantees that the packet will be delivered within a specified host-to-host delay bound.

In-Order packets: This service ensures that the packet arrives at the destination in the order in which they are sent.

Guaranteed max jitter: This service ensures that the amount of time taken between two successive transmissions at the sender is equal to the time between their receipt at the destination.

Security services: The network layer provides security by using a session key between the source and destination host. The network layer in the source host encrypts the payloads of datagrams being sent to the destination host.

The network layer in the destination host would then decrypt the payload.

IPv4 addresses

IPv4 addresses 32 bit binary addresses (divided into 4 octets) used by the Internet Protocol (OSI Layer 3) for delivering packet to a device located in same or remote network.

MAC address (Hardware address) is a globally unique address which represents the network card and cannot be changed.

IPv4 address refers to a logical address, which is a configurable address used to identify which network this host belongs to and also a network specific host number. In other words, an IPv4 address consists of two parts; a network part and a host part.

There are two main IPv4 address spaces—The public address space and the private address space.

The primary difference between both address spaces is that the public IPv4 addresses are routable on the internet, which means that any device that requires communication to other devices on the internet will need to be assigned a public IPv4 address on its interface, which is connected to the internet.

The public address space is divided into five classes:

Class A	0.0.0.0 – 126.255.255.255
Class B	128.0.0.0 – 191.255.255.255
Class C	192.0.0.0 – 223.255.255.255
Class D	224.0.0.0 – 239.255.255.255
Class E	240.0.0.0 – 255.255.255.255

Private IPv4 addresses

As defined by **RFC(Request For Comments) 1918**, there are three classes of private IPv4 address that are allocated for private use only. This means within a private network such as LAN.

The benefit of using the private address space (RFC 1918) is that the classes are not unique to any particular organization or group.

They can be used within an organization or a private network. However, on the internet, the public IPv4 address is unique to a device.

This means that if a device is directly connected to the internet with a private IPv4 address, there will be no network connectivity to devices on the internet.

Most ISPs usually have a filter to prevent any private addresses (RFC 1918) from entering their network.

Classful Addressing

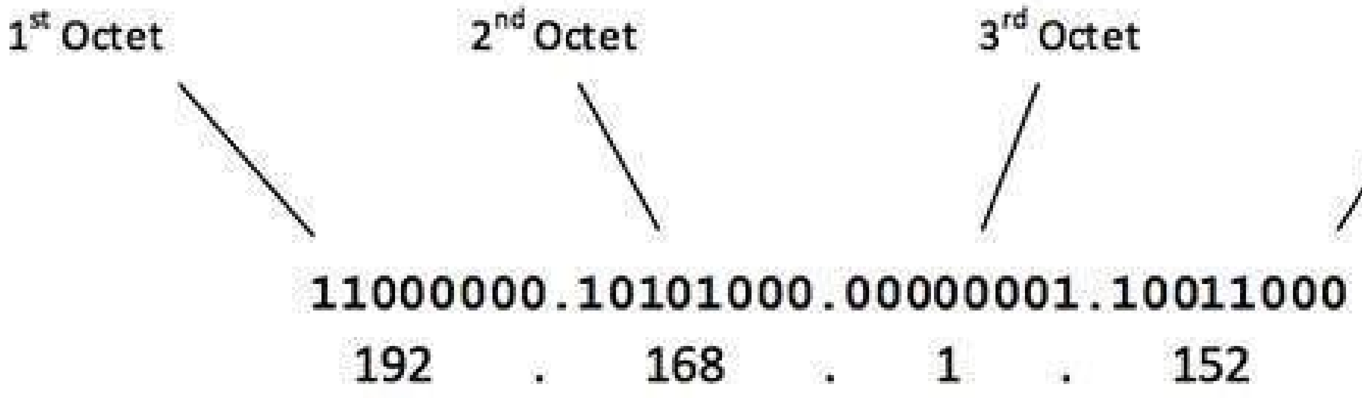
The 32-bit IP address is divided into five sub-classes. These are:
Class A
Class B
Class C
Class D
Class E

Each of these classes has a valid range of IP addresses. Classes D and E are reserved for multicast and experimental purposes, respectively.

The order of bits in the first octet determine the classes of IP address.

Internet Corporation for Assigned Names and Numbers is responsible for assigning IP addresses.

The first octet referred here is the left most of all. The octets numbered as follows depicting dotted decimal notation of IP Address −



The number of networks and the number of hosts per class can be derived by this formula −

Number of networks = 2^network_bits

Number of Hosts/Network = 2^host_bits - 2

When calculating hosts' IP addresses, 2 IP addresses are decreased because they cannot be assigned to hosts, i.e. the first IP of a network is network number and the last IP is reserved for Broadcast IP.

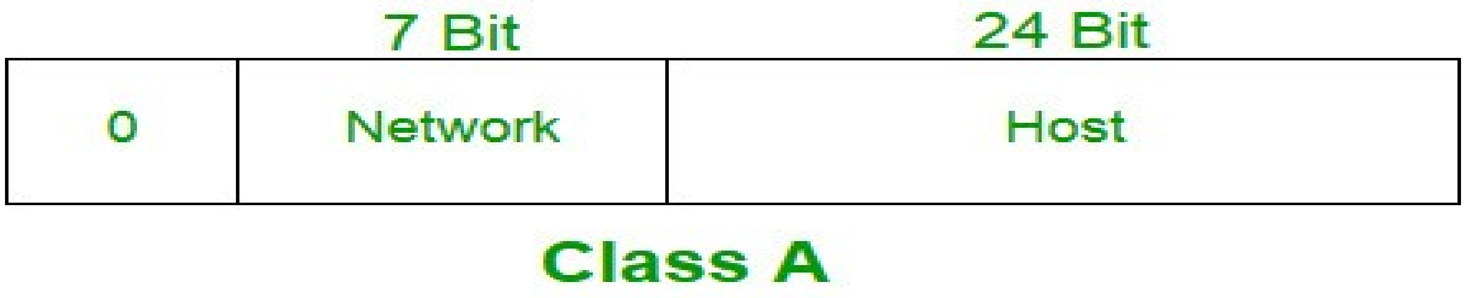
Class A Address

The first bit of the first octet is always set to 0 (zero). Thus, the first octet ranges from 1 – 127, i.e.

00000001 - 01111111
1 - 127

Class A addresses only include IP starting from 1.x.x.x to 126.x.x.x only. The IP range 127.x.x.x is reserved for loopback IP addresses.
The default subnet mask for Class A IP address is 255.0.0.0 which implies that Class A addressing can have 126 networks (2^7-2) and 16777214 hosts (2^{24}-2).
Class A IP address format is thus: **0NNNNNNN**.HHHHHHHH.HHHHHHHH.HHHHHHHH

2^7-2= 126 network ID(Here 2 address is subtracted because 0.0.0.0 and 127.x.y.z are special address.)
2^24 – 2 = 16,777,214 host ID
IP addresses belonging to class A ranges from 1.x.x.x – 126.x.x.x



Class B Address

An IP address which belongs to class B has the first two bits in the first octet set to 10, i.e.

10000000 - 10111111
128 - 191

Class B IP Addresses range from 128.0.x.x to 191.255.x.x. The default subnet mask for Class B is 255.255.x.x.
Class B has 16384 (2^{14}) Network addresses and 65534 (2^{16}-2) Host addresses.
Class B IP address format is: **10NNNNNN.NNNNNNNN**.HHHHHHHH.HHHHHHHH

2^14 = 16384 network address
2^16 – 2 = 65534 host address
IP addresses belonging to class B ranges from 128.0.x.x – 191.255.x.x.

Class B

Class C Address

The first octet of Class C IP address has its first 3 bits set to 110, that is −

11000000 - 11011111

192 - 223

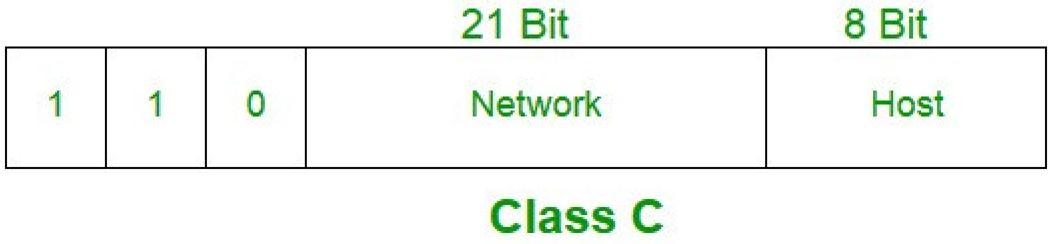
Class C IP addresses range from 192.0.0.x to 223.255.255.x. The default subnet mask for Class C is 255.255.255.x.

Class C gives 2097152 (2^{21}) Network addresses and 254 (2^{8}-2) Host addresses.

Class C IP address format is: **110NNNNN.NNNNNNNN.NNNNNNNN**.HHHHHHHH

2^21 = 2097152 network address

2^8 – 2 = 254 host address



Class D Address

Very first four bits of the first octet in Class D IP addresses are set to 1110, giving a range of

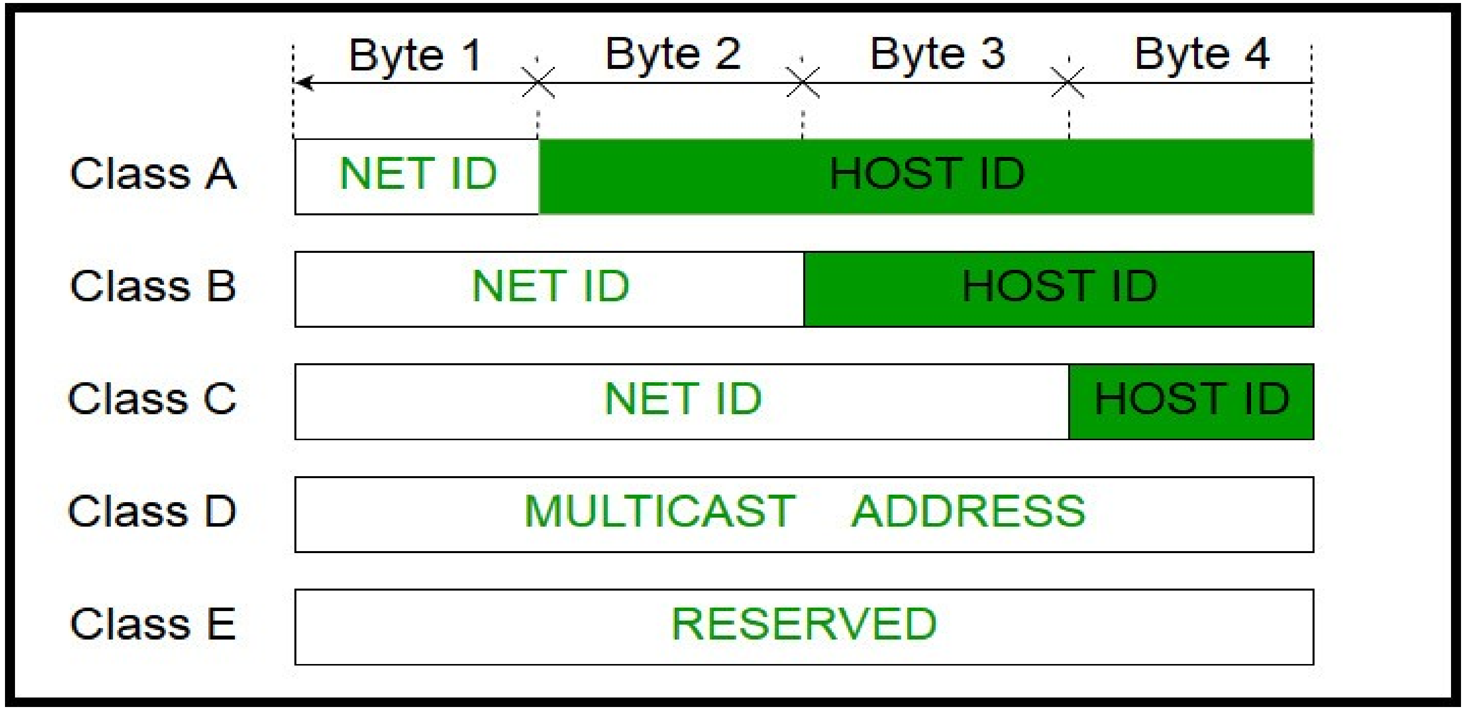
11100000 - 11101111

224 - 239

Class D has IP address range from 224.0.0.0 to 239.255.255.255. Class D is reserved for Multicasting. In multicasting data is not destined for a particular host, that is why there is no need to extract host address from the IP address, and Class D does not have any subnet mask.

Class E Address

This IP Class is reserved for experimental purposes only for R&D or Study. IP addresses in this class ranges from 240.0.0.0 to 255.255.255.254. Like Class D, this class too is not equipped with any subnet mask.



Problems with Classful Addressing:

The problem with this classful addressing method is that millions of class A address are wasted, many of the class B address are wasted, whereas number of addresses available in class C is so small that it cannot cater the needs of organizations.

Class D addresses are used for multicast routing and are therefore available as a single block only. Class E addresses are reserved.

Since there are these problems, Classful networking was replaced by Classless Inter-Domain Routing (CIDR) in 1993. We will be discussing Classless addressing in next post.

Classless Addressing-

Classless Addressing is an improved IP Addressing system.

It makes the allocation of IP Addresses more efficient.

It replaces the older classful addressing system based on classes.

It is also known as **Classless Inter Domain Routing (CIDR)**.

CIDR Block-

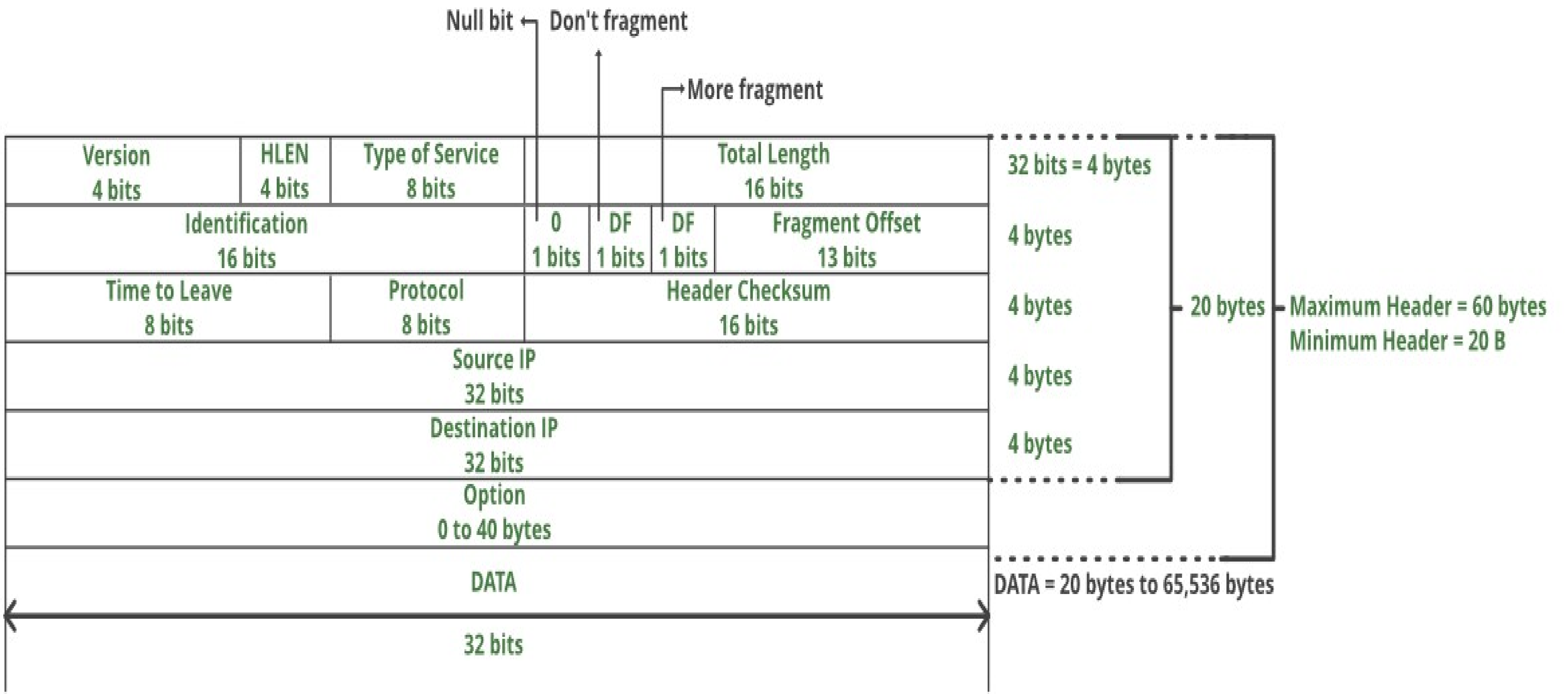
When a user asks for specific number of IP Addresses,

CIDR dynamically assigns a block of IP Addresses based on certain rules.

This block contains the required number of IP Addresses as demanded by the user.

This block of IP Addresses is called as a **CIDR block**.

IPV4 Header Format



VERSION: Version of the IP protocol (4 bits), which is 4 for IPv4

HLEN: IP header length (4 bits), which is the number of 32 bit words in the header. The minimum value for this field is 5 and the maximum is 15.

Type of service: Low Delay, High Throughput, Reliability (8 bits)

Total Length: Length of header + Data (16 bits), which has a minimum value 20 bytes and the maximum is 65,535 bytes.

Identification: Unique Packet Id for identifying the group of fragments of a single IP datagram (16 bits)

Flags: 3 flags of 1 bit each : reserved bit (must be zero), do not fragment flag, more fragments flag (same order)

Fragment Offset: Represents the number of Data Bytes ahead of the particular fragment in the Datagram. Specified in terms of number of 8 bytes, which has the maximum value of 65,528 bytes.

Time to live: Datagram's lifetime (8 bits), It prevents the datagram to loop through the network by restricting the number of Hops taken by a Packet before delivering to the Destination.

Protocol: Name of the protocol to which the data is to be passed (8 bits)

Header Checksum: 16 bits header checksum for checking errors in the datagram header

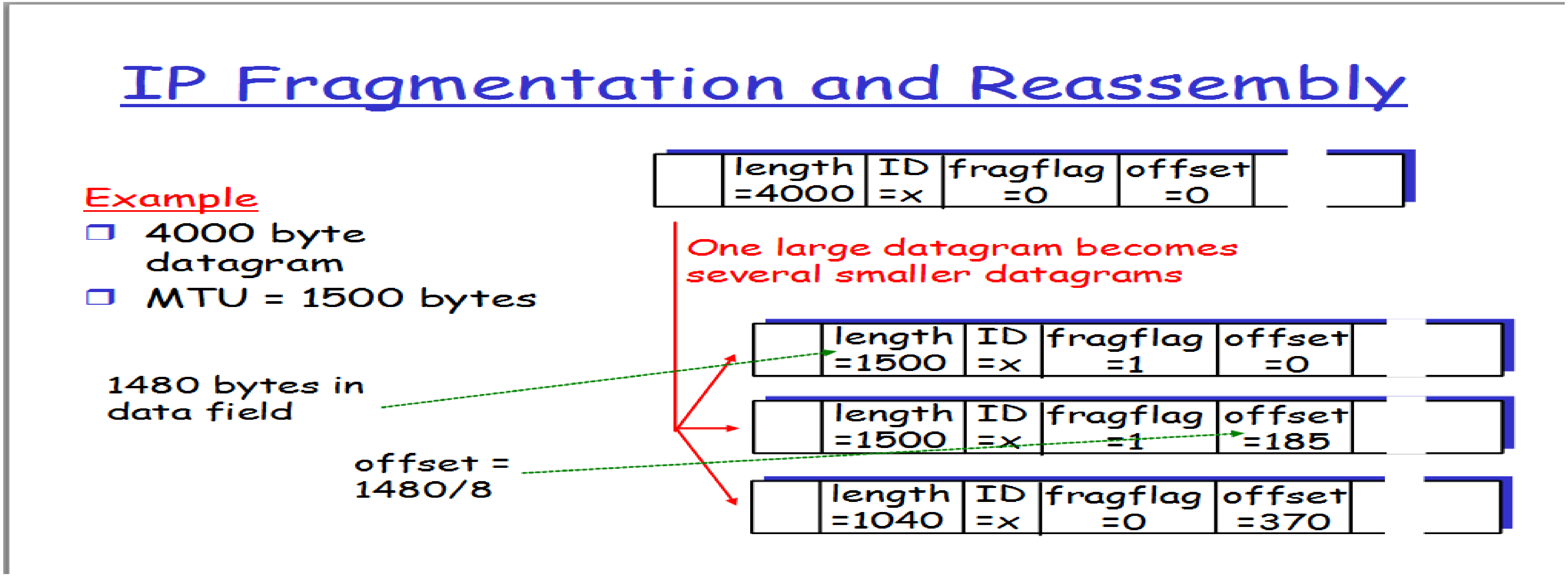
Source IP address: 32 bits IP address of the sender

Destination IP address: 32 bits IP address of the receiver

Option: Optional information such as source route, record route. Used by the Network administrator to check whether a path is working or not.

IP Fragmentation

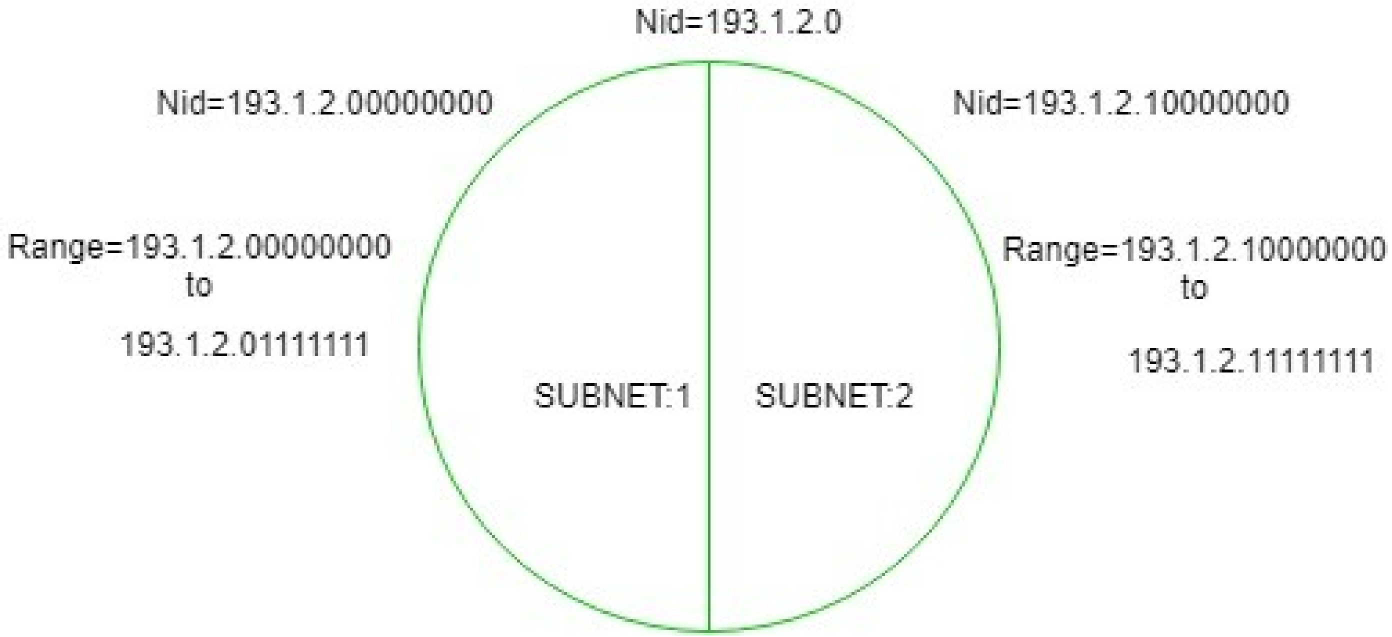
Fragmentation is done by the network layer when the maximum size of datagram is greater than maximum size of data that can be held a frame i.e., its Maximum Transmission Unit (MTU). The network layer divides the datagram received from transport layer into fragments so that data flow is not disrupted.



Subnetting

When a bigger network is divided into smaller networks, in order to maintain security, then that is known as Subnetting. so, maintenance is easier for smaller networks.

Now, let's talk about dividing a network into two parts:
so, to divide a network into two parts, you need to choose one bit for each Subnet from the host ID part.



In the above diagram, there are two Subnets.

Note: It is a class C IP so, there are 24 bits in the network id part and 8 bits in the host id part.

For Subnet-1:

The first bit which is chosen from the host id part is zero and the range will be from (193.1.2.00000000 till you get all 1's in the host ID part i.e., 193.1.2.01111111) except for the first bit which is chosen zero for subnet id part. Thus, the range of subnet-1:

193.1.2.0 to 193.1.2.127

For Subnet-2:

The first bit chosen from the host id part is one and the range will be from (193.1.2.100000000 till you get all 1's in the host ID part i.e, 193.1.2.11111111).Thus, the range of subnet-2:

193.1.2.128 to 193.1.2.255

Network mask & Masking

Masking is process that extracts the address of the physical network form an IP address. Masking can be done whether we have subnetting or not. If we have not subnetted the network, masking extracts the network address form an IP address. If we have subnetted, masking extracts the subnetwork address form an IP address.

A **Netmask** is a 32-bit "mask" used to divide an IP address into subnets and specify the network's available hosts.

In a netmask, two bits are always automatically assigned. For example, in 255.255.225.0, "0" is the assigned network address. In 255.255.255.255, "255" is the assigned broadcast address. The 0 and 255 are always assigned and cannot be used.

Types of Masking

There are two types of masking as given below

Boundary Level Masking:

If the masking is at the boundary level (the mask numbers are either 255 or 0), finding the subnetwork address is very easy. Follow these 2 rules

- The bytes in IP address that correspond to 255 in the mask will be repeated in the subnetwork address.

- The bytes in IP address that correspond to 0 in the mask will change to 0 in the subnetwork address.

e.g.,		
	IP address	45 . 23 . 21 . 8
	Mask	255 . 255 . 0 . 0
	Subnetwork address	45 . 23 . 0 . 0

Non-boundary Level Masking: If the masking is not at the boundary level (the mask numbers are not just 255 or 0), finding subnetwork address involves using the bit-wise AND operator, follow these 3 rules

- The bytes in IP address that correspond to 255 in the mask will be repeated in the subnetwork address.
- The bytes in the IP address that correspond to 0 in the mask will be change to 0 in the subnetwork address.
- For other bytes, use the bit-wise AND operator

e.g.,

	IP address	213 . 23 . 47 . 37
	Mask	255 . 255 . 255 . 240
	Subnetwork address	213 . 23 . 47 . 32

As we can see, 3 bytes are ease {, to determine. However, the 4th bytes needs the bit-wise AND operation.

Example:- what is the subnetwork address if the destination address is 200.45.34.56 and subnet mask is 255.255.240.0 ?

Solution:- Using AND operation , we find subnet wok address

1) Convert the given destination address to binary format

Given destination Address 200.45.34.56 11001000.00101101.00100010.00111000

2) Convert the given subnet mask to binary format

Subnet mask Address 255.255.240.0 11111111.11111111.11110000.00000000

3) Do the **AND** operation

The subnetwork address for the given is **200.45.32.0**

Given Destination Address	200.45.34.56	**11001000.00101101.00100010.00111000**
Subnet Mask Address	255.255.240.0	**11111111.11111111.11110000.00000000**
Subnetwork Address	200.45.32.0	**11001000.00101101.00100000.00000000**

Example:- for a network address 192.168.10.0 and subnet mask 255.255.255.224 then calculate number of subnet and number of host

Solution:- given network address 192.168.10.0 is class C address. Subnet address is 255.255.255.224. here three bits are borrowed for subnet.

1)Number of subnet and number of host

255.255.255.240 convert into binary -> 11111111.11111111.11111111.11100000

Number of subnet = 2^x $=2^3$ =8

Number of Host per subnet $=2^y - 2$ $=2^5 -2$ = 30

Network Address Translation (NAT)

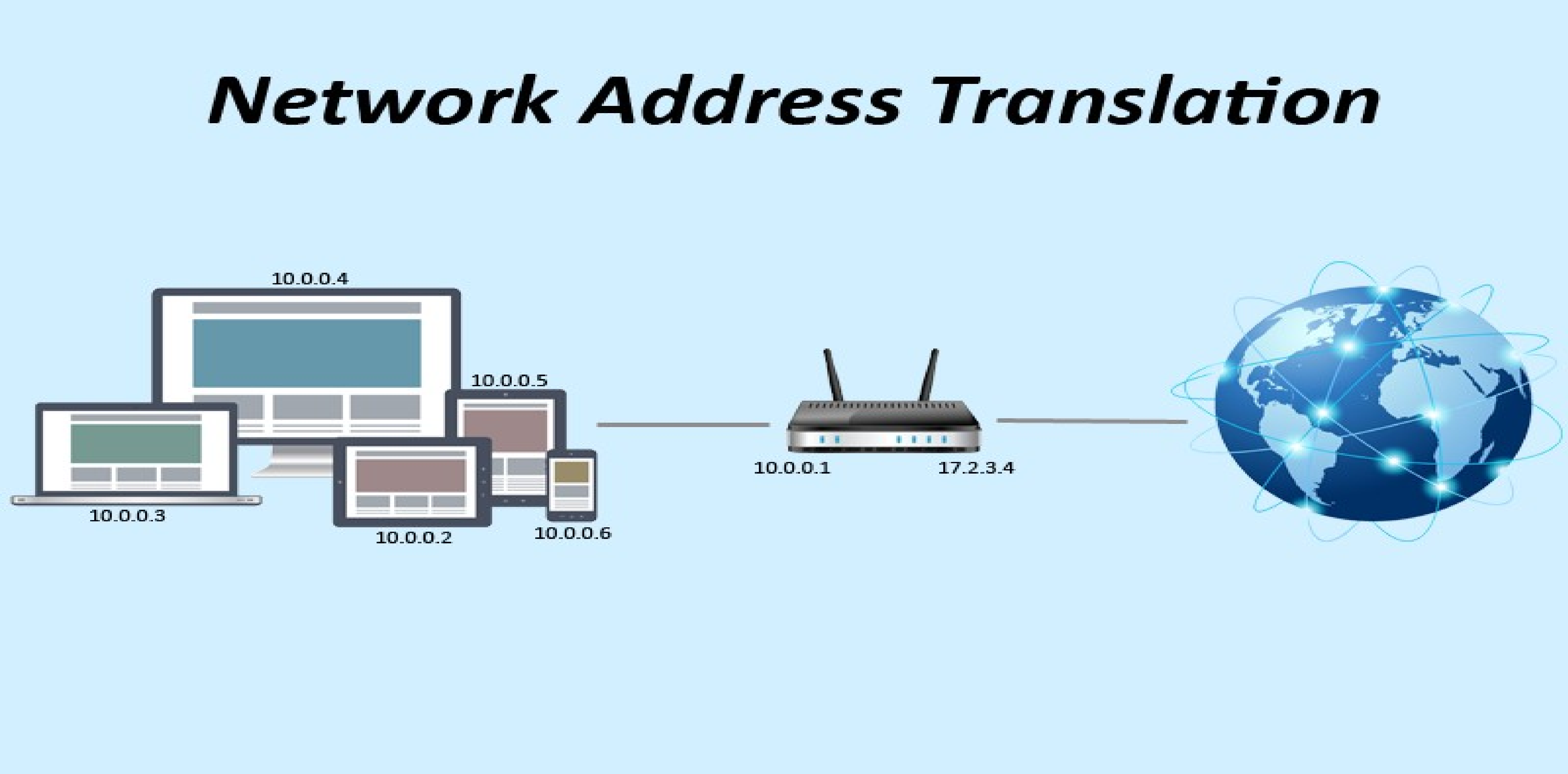
Access the Internet, one public IP address is needed, but we can use a private IP address in our private network.

The idea of NAT is to allow multiple devices to access the Internet through a single public address.

To achieve this, the translation of private IP address to a public IP address is required.

Network Address Translation (NAT) is a process in which one or more local IP address is translated into one or more Global IP address and vice versa in order to provide Internet access to the local hosts. Also, it does the translation of port numbers i.e. masks the port number of the host with another port number, in the packet that will be routed to the destination.

It then makes the corresponding entries of IP address and port number in the NAT table. NAT generally operates on router or firewall.


Network Address Translation
10.0.0.4
10.0.0.5
10.0.0.3
10.0.0.2
10.0.0.6
10.0.0.1
17.2.3.4

Supernetting in Network Layer

Supernetting is the opposite of Subnetting. In subnetting, a single big network is divided into multiple smaller subnetworks.

- In Supernetting, multiple networks are combined into a bigger network termed as a Super network or Supernet.
- Supernetting is mainly used in Route Summarization, where routes to multiple networks with similar network prefixes are combined into a single routing entry, with the routing entry pointing to a Super network, encompassing all the networks.
- This in turn significantly reduces the size of routing tables and also the size of routing updates exchanged by routing protocols.

More specifically,

When multiple networks are combined to form a bigger network, it is termed as super-netting

Super netting is used in route aggregation to reduce the size of routing tables and routing table updates

There are some points which should be kept in mind while supernetting:

- All the Networks should be contiguous.
- The block size of every networks should be equal and must be in form of 2^n.
- First Network id should be exactly divisible by whole size of supernet.

Example – Suppose 4 small networks of class C:
200.1.0.0, 200.1.1.0, 200.1.2.0, 200.1.3.0
Build a bigger network which have a single Network Id.
Explanation – Before Supernetting routing table will be look like as:

NETWORK ID	SUBNET MASK	INTERFACE
200.1.0.0	255.255.255.0	A
200.1.1.0	255.255.255.0	B
200.1.2.0	255.255.255.0	C
200.1.3.0	255.255.255.0	D

First, lets check whether three condition are satisfied or not:
Contiguous: You can easily see that all network are contiguous all having size 256 hosts.
Range of first Network from 200.1.0.0 to 200.1.0.255. If you add 1 in last IP address of first network that is 200.1.0.255 + 0.0.0.1, you will get the next network id that is 200.1.1.0. Similarly, check that all network are contiguous.

Equal size of all network: As all networks are of class C, so all of the have a size of 256 which in turn equal to 2^8.

First IP address exactly divisible by total size: When a binary number is divided by 2^n then last n bits are the remainder.

Hence in order to prove that first IP address is exactly divisible by while size of Supernet Network.

You can check that if last n bits are 0 or not. In given example first IP is 200.1.0.0 and whole size of supernet is $4*2^8 = 2^{10}$. If last 10 bits of first IP address are zero then IP will be divisible.

11001000	00000001	00000000	00000000
200 .	1 .	0 .	0

Last 10 bits of first IP address are zero (highlighted by green color). So 3rd condition is also satisfied.

Therefore, you can join all these 4 networks and can make a Supernet. New Supernet Id will be 200.1.0.0.

Advantages of Supernetting –

Control and reduce network traffic

Helpful to solve the problem of lacking IP addresses

Minimizes the routing table

Disadvantages of Supernetting –

It cannot cover different area of network when combined

All the networks should be in same class and all IP should be contiguous

FORWARDING IP Packets:-

Forwarding means to place the packet in its route to its destination.

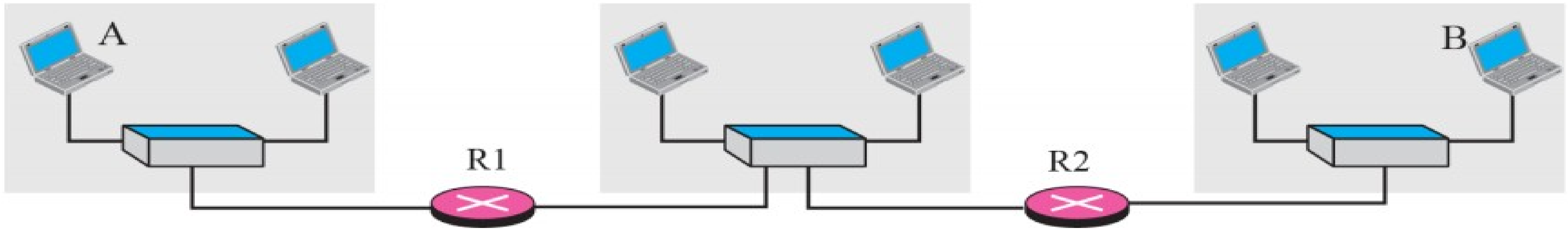
Since the Internet today is made of a combination of links (networks), forwarding means to deliver the packet to the next hop (which can be the final destination or the intermediate connecting device).

Although the IP protocol was originally designed as a connectionless protocol, today the tendency is to use IP as a connection-oriented protocol.

Forwarding Techniques:

- Next-Hop Method
- Network-Specific Method
- Host-Specific Method
- Default Method

Next-Hop Routing – Table only holds the address of the next hop (instead info regarding the entire route)



A

Destination	Route
Host B	R1, R2, Host B

R1

Destination	Route
Host B	R2, Host B

R2

Destination	Route
Host B	Host B

a. Routing tables based on route

A

Destination	Next Hop
Host B	R1

R1

Destination	Next Hop
Host B	R2

R2

Destination	Next Hop
Host B	---

b. Routing tables based on next hop

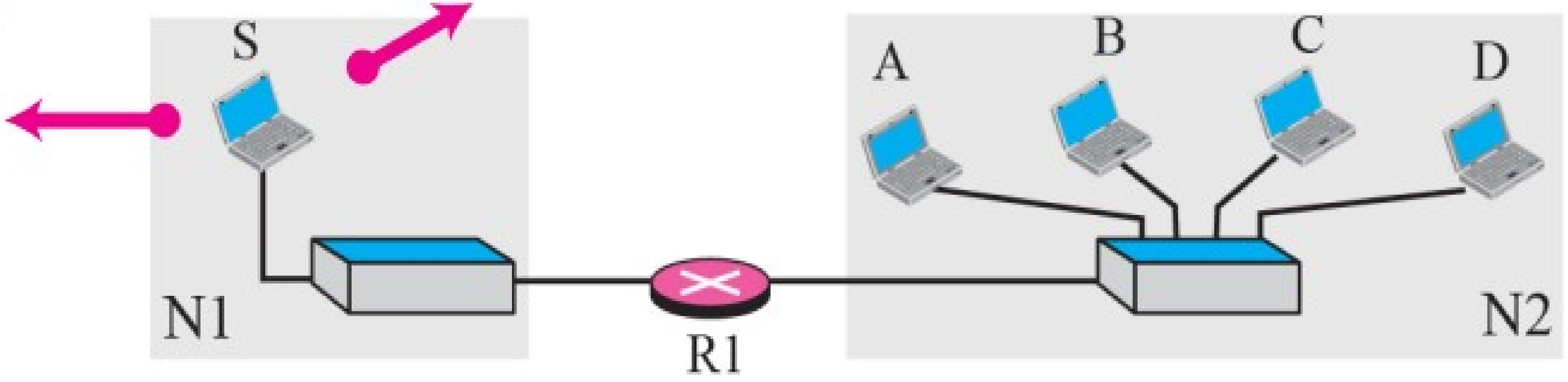
Network-Specific Routing – instead of an entry for each host on the same physical network, only one entry for the network is defined

Network-specific routing table for host S

Destination	Next Hop
N2	R1

Host-specific routing table for host S

Destination	Next Hop
A	R1
B	R1
C	R1
D	R1



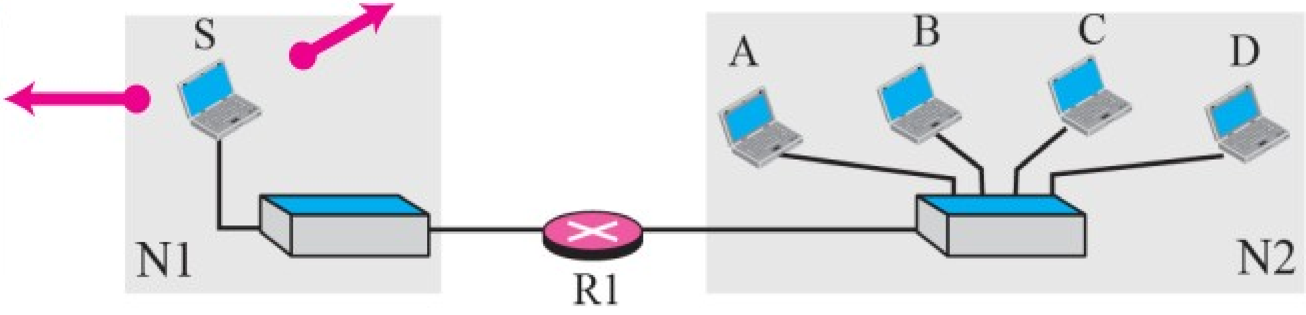
Host-Specific Routing – for a specific destination host, you might want to control the exact route – in this case, the actual Rx is listed in the routing table and the desired next hop is listed

Network-specific routing table for host S

Destination	Next Hop
N2	R1

Host-specific routing table for host S

Destination	Next Hop
A	R1
B	R1
C	R1
D	R1

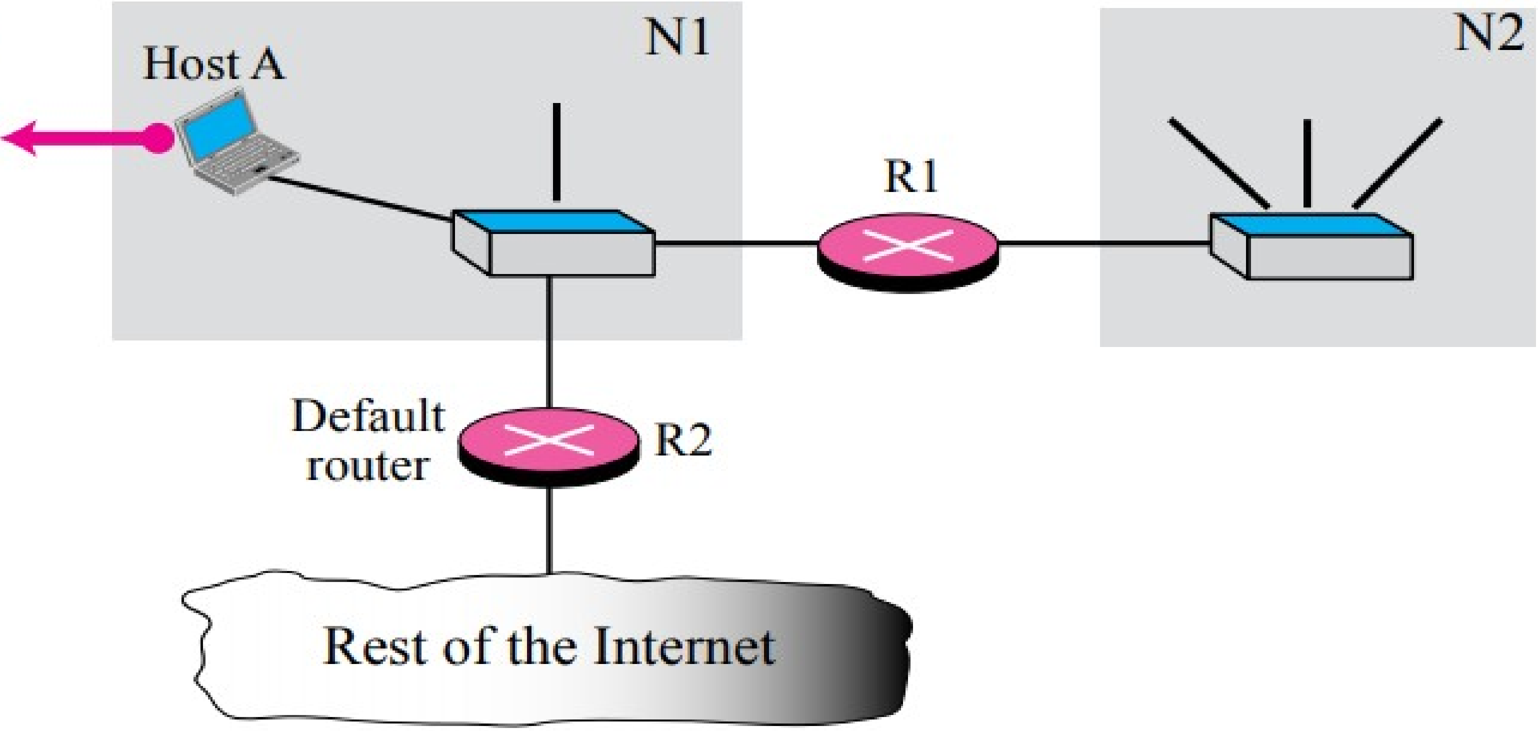


In this case, you want every packet traveling to Host B to traverse through R3. For the other hosts on N2 and N3, the Network-specific routing approach is used.

Default Routing – instead of listing all of the various networks in the Internet, Tx host would use one entry called the Default (network address 0.0.0.0)

Routing table for host A

Destination	Next Hop
N2	R1
......	......
Default	R2



In this case, R1 sends to a specific network however, R2 sends to the remainder of the Internet (default)

Internet Protocol version 6 (IPv6)

IPv6 was developed by Internet Engineering Task Force (IETF) to deal with the problem of IP v4 exhaustion.

IP v6 is 128-bits address having an address space of 2^128, which is way bigger than IPv4.

In IPv6 we use Colon-Hexa representation. There are 8 groups and each group represents 2 Bytes.

Advantages of IPV6

1)Larger address space
2)Better header format
3)Security capabilities
4)Support for resource allocation
5)New options
6)Allowance for extension

Disadvantages of IPv6

Conversion: Due to widespread present usage of IPv4 it will take a long period to completely shift to IPv6.

Communication: IPv4 and IPv6 machines cannot communicate directly with each other. They need an intermediate technology to make that possible.

Types of IPv6 Address

Now that we know about what is IPv6 address let's take a look at its different types.

Unicast addresses

It identifies a unique node on a network and usually refers to a single sender or a single receiver.

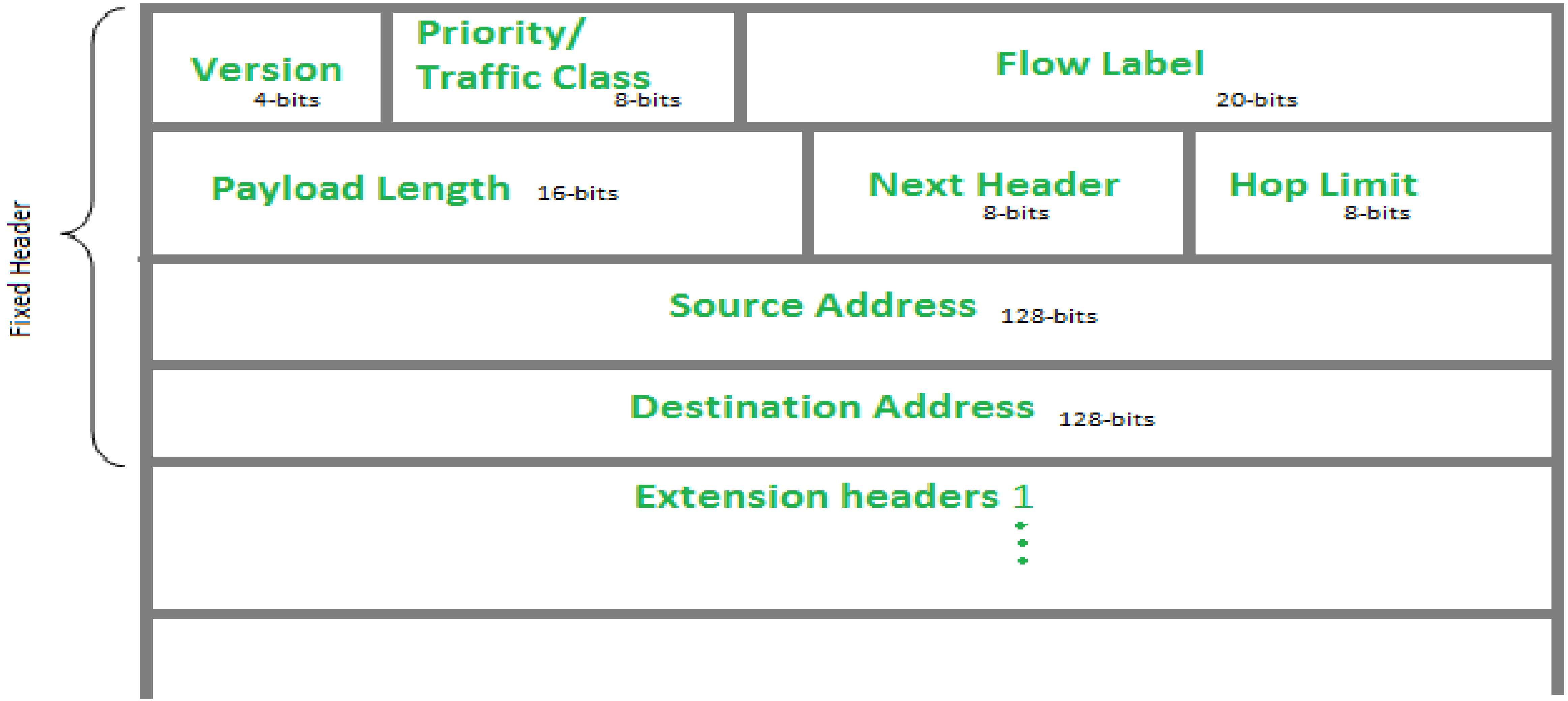
Multicast addresses

It represents a group of IP devices and can only be used as the destination of a datagram.

Anycast addresses

It is assigned to a set of interfaces that typically belong to different nodes

IP version 6 Header Format :



Version (4-bits) : Indicates version of Internet Protocol which contains bit sequence 0110.

Traffic Class (8-bits) : Traffic Class (8-bits): These 8 bits are divided into two parts. The most significant 6 bits are used for Type of Service to let the Router Known what services should be provided to this packet. The least significant 2 bits are used for Explicit Congestion Notification (ECN).

As of now only 4-bits are being used (and remaining bits are under research), in which 0 to 7 are assigned to Congestion controlled traffic and 8 to 15 are assigned to Uncontrolled traffic.

Priority assignment of Congestion controlled traffic :

Priority	Meaning
0	No Specific traffic
1	Background data
2	Unattended data traffic
3	Reserved
4	Attended bulk data traffic
5	Reserved
6	Interactive traffic
7	Control traffic

Flow Label (20-bits) : This label is used to maintain the sequential flow of the packets belonging to a communication. The source labels the sequence to help the router identify that a particular packet belongs to a specific flow of information. This field helps avoid re-ordering of data packets. It is designed for streaming/real-time media.

Payload Length (16-bits) : It is a 16-bit (unsigned integer) field, indicates total size of the payload which tells routers about amount of information a particular packet contains in its payload. Payload Length field includes extension headers (if any) and upper layer packet.

Next Header (8-bits) : Next Header indicates type of extension header(if present) immediately following the IPv6 header. Whereas In some cases it indicates the protocols contained within upper-layer packet, such as TCP, UDP.

Hop Limit (8-bits) : Hop Limit field is same as TTL in IPv4 packets. It indicates the maximum number of intermediate nodes IPv6 packet is allowed to travel. Its value gets decremented by one, by each node that forwards the packet and packet is discarded if value decrements to 0.

Source Address (128-bits) : Source Address is 128-bit IPv6 address of the original source of the packet.

Destination Address (128-bits) : Destination Address field indicates the IPv6 address of the final destination(in most cases). All the intermediate nodes can use this information in order to correctly route the packet.

Extension Headers : In order to rectify the limitations of *IPv4 Option Field*, Extension Headers are introduced in IPversion 6. The extension header mechanism is very important part of the IPv6 architecture

Transition from IPv4 to IPv6 address

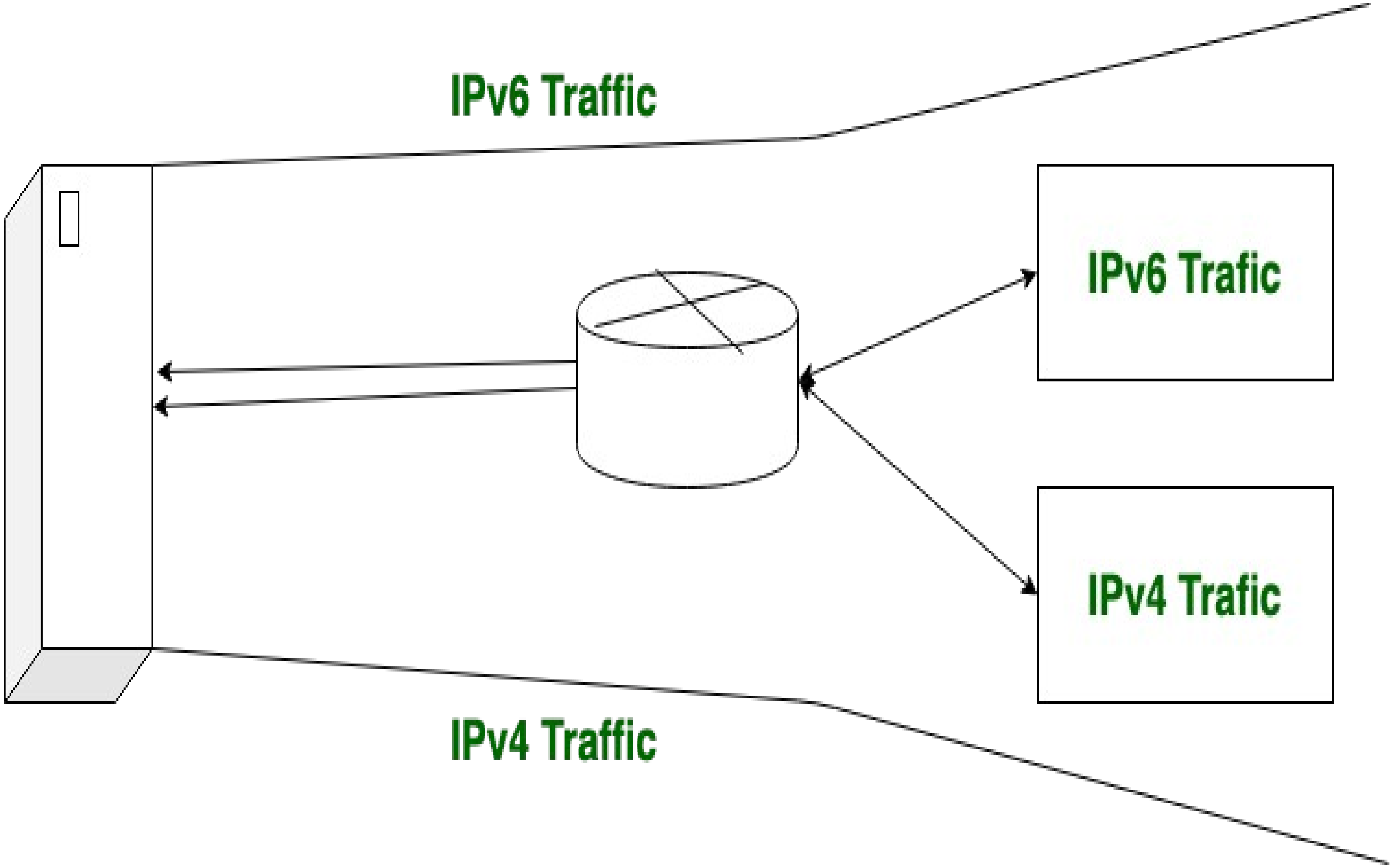
When we want to send a request from an IPv4 address to an IPv6 address but it isn't possible because IPv4 and IPv6 transition is not compatible. For solution to this problem, we use some technologies.

These technologies are: *Dual Stack Routers, Tunneling, and NAT Protocol Translation*. These are explained as following below.

Dual Stack Routers:

In dual stack router, A router's interface is attached with Ipv4 and IPv6 addresses configured is used in order to transition from IPv4 to IPv6.

In this above diagram, A given server with both IPv4 and IPv6 address configured can communicate with all hosts of IPv4 and IPv6 via dual stack router (DSR). The dual stack router (DSR) gives the path for all the hosts to communicate with server without changing their IP addresses.

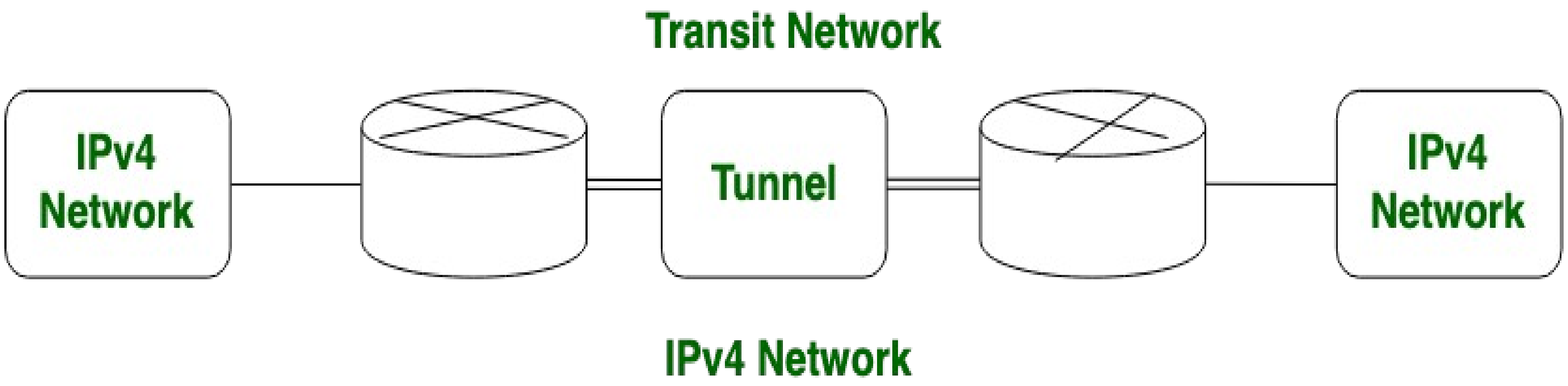

IPv6 Traffic
IPv6 Trafic
IPv4 Trafic
IPv4 Traffic

Tunneling:

Tunneling is used as a medium to communicate the transit network with the different IP versions. In this above diagram, the different IP versions such as IPv4 and IPv6 are present.

The IPv4 networks can communicate with the transit or intermediate network on IPv6 with the help of Tunnel.

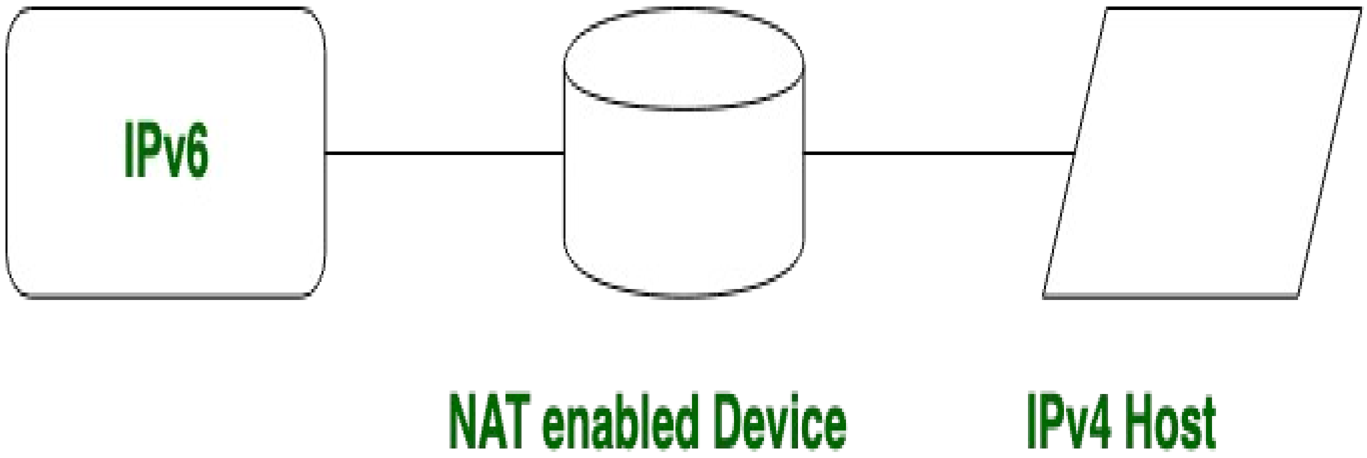
Its also possible that the IPv6 network can also communicate with IPv4 networks with the help of Tunnel.



NAT Protocol Translation:

By the help of NAT Protocol Translation technique, the IPv4 and IPv6 networks can also communicate with each other which do not understand the address of different IP version. Generally, an IP version doesn't understand the address of different IP version, for the solution of this problem we use NAT-PT device which remove the header of first (sender) IP version address and add the second (receiver) IP version address so that the Receiver IP version address understand that the request is send by the same IP version, and its vice-versa is also possible.

In above diagram, an IPv4 address communicate with the IPv6 address via NAT-PT device to communicate easily. In this situation IPv6 address understand that the request is send by the same IP version (IPv6) and it respond.



Difference Between IPv4 and IPv6:

IPV4	IPV6
IPv4 has 32-bit address length	IPv6 has 128-bit address length
It Supports Manual and DHCP address configuration	It supports Auto and renumbering address configuration
In IPv4 end to end connection integrity is Unachievable	In IPv6 end to end connection integrity is Achievable
It can generate 4.29×109 address space	Address space of IPv6 is quite large it can produce 3.4×1038 address space
Security feature is dependent on application	IPSEC is inbuilt security feature in the IPv6 protocol
Address representation of IPv4 in decimal	Address Representation of IPv6 is in hexadecimal
Fragmentation performed by Sender and forwarding routers	In IPv6 fragmentation performed only by sender
In IPv4 Packet flow identification is not available	In IPv6 packetflow identification are Available and uses flow label field in the header
In IPv4 checksumfield is available	In IPv6 checksumfield is not available
It has broadcast Message Transmission Scheme	In IPv6 multicast and any cast message transmission scheme is available